



# OPEN

Compute Project

## Diablo 400 Project: Rack and Power

0.5.2

**Base Specification**  
Effective May 30<sup>th</sup>, 2025

Authors: Microsoft, Meta, Google

## 1 Version History

| Date        | Version # | Authors                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Description                                                                           |
|-------------|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| 5 May 2025  | 0.5.0     | <b>Microsoft:</b> Laurentiu Olariu, Samir Kala, Joel Jaramillo, Kelsey Wildstone, Brandon DeVaul, Harsha Bojja, Andy Regimbal, Jason Adrian.<br><br><b>Meta:</b> David Sun, Dmitriy Shapiro, Ben Kim, John Fernandes, Drazena Brocilo, Mingchun Xu.<br><br><b>Google:</b> Jorge Rodriguez, Xiong Li, Christina Peabody, Vijay Boovaragavan, Matt Tamashiro, Bill Barry, Maziar Mobarrez, Adrian Shanov, Scott McCauley, Tamar Makharashvili, Vivek Khaire | Initial Release                                                                       |
| May 2025    | 0.5.1     |                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Updated Authors                                                                       |
| 30 May 2025 | 0.5.2     | Same as above                                                                                                                                                                                                                                                                                                                                                                                                                                             | Updated Interface Section and Compliance Section. Updated PSU power for 400Vac input. |
|             |           |                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                       |
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## 5 Acknowledgements

The Contributors of this Specification would like to acknowledge the following for their feedback:

**Microsoft:** Laurentiu Olariu, Samir Kala, Joel Jaramillo, Kelsey Wildstone, Brandon DeVaul, Harsha Bojja, Jason Adrian, Andy Regimbal

**Meta:** David Sun, Dmitriy Shapiro, Ben Kim, John Fernandes, Drazena Brocilo, Mingchun Xu.

**Google:** Jorge Rodriguez, Xiong Li, Christina Peabody, Vijay Boovaragavan, Matt Tamashiro, Bill Barry, Maziar Mobarrez, Adrian Shanov, Scott McCauley, Tamar Makharashvili.

## **6 Compliance with OCP Tenets**

### **6.1 Openness**

This specification will hopefully serve as the future fundamental base for a disaggregated power rack to deliver  $\pm 400V_{DC}$  to a nearby IT rack. The base of the mechanical rack follows the Open Rack standard. Currently three companies have worked together to provide a high-level overview of the Diablo 400V architecture. The goal is to standardize items such as, high voltage connectors and mating interfaces.

### **6.2 Efficiency**

Transferring power at a higher voltage is more efficient, reduces I<sup>2</sup>R losses. This disaggregated power rack can be used with multiple generations and SKUs of IT racks; a disaggregated power rack can remain in the same location in the datacenter while IT racks can be swapped as required. This will ultimately reduce the overall infrastructure costs. One objective is to use learnings from this architecture to distribute high direct current voltage in the datacenter directly to the IT racks to improve overall end to end power distribution efficiency.

### **6.3 Impact**

This is the first disaggregated power rack specification. Standardizing the high voltage rack architecture will hopefully lead to a common supply chain, provide end to end power efficiency gains, and leverage mainstream EV industry technologies in HVDC charging/battery infrastructure.

### **6.4 Scale**

The fundamental advantage of a disaggregated power rack is scalability in terms of rated power, redundancy schemes and backup power.

### **6.5 Sustainability**

Disaggregated power rack has the ability to be scaled for various power ratings allowing it to be re-used for multiple generations of an IT rack. The IT rack can be updated while maintaining the same hardware in the disaggregated power rack, therefore reducing the need to scrap power building blocks.



## 8 Rack Physical Dimensions

Table 1 included below illustrates the Diablo 400 power rack dimensions. The height of the rack with casters is not included in the table below. The target loading capacity of the rack is 6000 lbs.

**Table 1:** Various rack dimensions

|           | Height (mm)  | Width (mm)  | Depth (mm)   |
|-----------|--------------|-------------|--------------|
| Microsoft | 2329 $\pm$ 3 | 600 $\pm$ 2 | 1200 $\pm$ 2 |
| Meta      | 2329         | 600         | 1219         |
| Google    | 2329*        | 711         | 1219.2       |

\*Option for height up to 2410mm

### 8.1 Mechanical Features

- Sliding shelves to facilitate push in/out of big building blocks such as a 4OU BBU, etc.
  - Blind mate connector with static rail/sliding rails for PSU/BBU/CBU
  - Center justified busbar placement
- Safety doors per compliance chapter 9
  - Provision to add a locking door on both the front and back of the rack, full or partial doors
  - Provision for a switch for intrusion
- Reserved space for structural support
- Connection for AC Cables (Lug vs Connector)
  - Option for connectors instead of lugs
  - Example - Quick Connect (Amphenol Surelock) connector
- Cable routing
  - Options to support AC whips routing
  - Option for AC inlet in the back of the top of the rack, DC output on the top of the front of the rack
- Accessibility defined per safety hazard in chapter 9

## 9 Voltage Distributions

This section describes AC input voltage and DC output voltage requirements as well as various energy storage options.

### 9.1 Rack AC Power Requirements

#### 9.1.1 AC Input Voltage

The rack is designed to accept 3-phase AC input (3P+E) voltage into an AC PDU, which then distributes the AC power as input to the AC/DC PSU modules.

The rack is configurable to support both single 3-phase AC input feed or dual AC input for N+N redundancy feed.

**Table 2:** Various AC input option configurations.

| Parameter                      | Nominal1             | Nominal2             | Nominal3             | Min/Max |
|--------------------------------|----------------------|----------------------|----------------------|---------|
| <b>WYE 5Wire<br/>3L+N+PE</b>   | 480 V <sub>rms</sub> | 415 V <sub>rms</sub> | 400 V <sub>rms</sub> | ±10%    |
| <b>Line to<br/>Neutral L+N</b> | 277 V <sub>rms</sub> | 240 V <sub>rms</sub> | 230 V <sub>rms</sub> | ±10%    |
| <b>Delta 4 Wire<br/>3L+PE</b>  | 480 V <sub>rms</sub> | 415 V <sub>rms</sub> | 400 V <sub>rms</sub> | ± 10%   |
| <b>Frequency</b>               | 50/60 Hz             | 50/60 Hz             | 50/60 Hz             | ±3Hz    |

#### 9.1.2 AC Input PF and THD

The power quality metrics, such as power factor (PF) and current harmonic distortion are listed in the table below.

**Table 3:** Power quality requirements.

| Parameter           | @480V <sub>AC</sub> Vin | @415V <sub>AC</sub> Vin | @400V <sub>AC</sub> Vin |
|---------------------|-------------------------|-------------------------|-------------------------|
| PF load <20%        | >0.95                   | >0.95                   | >0.95                   |
| PF load ≥20%        | >0.99                   | >0.99                   | >0.99                   |
| THD 5% < load <15%  | <15%                    | <15%                    | <15%                    |
| THD 15% ≤ load <30% | <10%                    | <10%                    | <10%                    |
| THD load ≥30%       | <5%                     | <5%                     | <5%                     |

### 9.1.3 AC Loss (Holdup and ride-trough)

Holdup notion applies to built-in PSU bulk storage that allows time until other form of energy support is engaged and activated to support the rack functions in case of AC loss. Separate from holdup, the ride-through capability of the PSU defines running without interruptions during voltage dropouts and sags/swells.

#### 9.1.3.1 PSU ride-through without BBU

Line Overvoltage: Follow ITIC curve for overvoltage and requires PSU to ride-through and not latch-off.

Line Sag: Follow both ITIC and IEC 61000-4-34 and requires PSU to ride-through and not latch-off.

Minimum holdup without BBU requirement is up to 20msec at 100% loading. Addition of system distributed holdup is acceptable, between the AC/DC PSU in the Diablo 400 rack and with the downstream DC/DC converter, located outside the rack.

#### 9.1.3.2 PSU ride-through with BBU

Shall ride-through any duration within the BBU hold up time range.

The PSU/Shelf hold-up energy/time shall be enough covering the smooth transition from AC drop out to BBU ramp up at the worst-case condition, which can be the AC drop at the lowest point of bulk voltage during peak load as defined in 9.3.2.

Holdup with BBU requirement shall be decided per solution specific requirements, sufficient to cover transition time.

## 9.2 Energy Storage Solutions

The rack is designed to support various storage solutions such as battery backup unit (BBU) and capacitor bank energy storage units (CBU). Depending on the user's requirement the number of BBUs and CBUs can be adjusted.

### 9.2.1 BBU Option

Rack shall support BBU options. Design shall support power ranges per electrical specification.

- Optional active management via microcontroller for secure access
- Optional module interface with a Rack Management Unit -interface standard defined in chapter 11 (CANBus, Ethernet, etc.)

### 9.2.1.1 Battery Backup Time

If BBU option is utilized, battery backup time shall support a range of 45 to 90 seconds holdup at 100% loading.

### 9.2.2 Capacitor Bank Option CBU

The rack design shall support capacitive bank energy storage, although power and duration may be specific to each application, and selection may differ per platform. Design shall support power ranges per electrical specification.

- Optional active management via microcontroller for secure access
- Optional module interface with a Rack Management Unit -interface standard defined in chapter TBD (CANBus, Ethernet, etc.)

## 9.3 Rack DC Power Requirements

### 9.3.1 DC Output Voltage for Nominal $\pm 400\text{VDC}$

DC output regulation table below. COM means common wire (neutral polarity between the  $+400\text{V}_{\text{DC}}$  and  $-400\text{V}_{\text{DC}}$  output. Voltages are measured at the output connector of the AC/DC PSU module or/and power shelf rectifier.

**Table 4:** DC output voltage parameters

| Parameter                                     | Nominal at no load         | Nominal at full load       | Min/Max                                                        |
|-----------------------------------------------|----------------------------|----------------------------|----------------------------------------------------------------|
| +400 to COM                                   | $410\text{V}_{\text{DC}}$  | $400\text{V}_{\text{DC}}$  | $\pm 0.5\%$                                                    |
| -400 to COM                                   | $-410\text{V}_{\text{DC}}$ | $-400\text{V}_{\text{DC}}$ | $\pm 0.5\%$                                                    |
| Ripple and noise                              |                            |                            | $\pm 1\%$ or $\pm 2\%$ pk_pk (maybe)                           |
| V_droop dc                                    |                            |                            | built in the Nominal setpoint ( $10\text{V} \pm 0.5\text{V}$ ) |
| Output voltage transient overshoot/undershoot |                            |                            | $\pm 3\%$ from typical set point                               |

### 9.3.2 DC Output Power for Nominal $\pm 400\text{VDC}$

The rack shall be capable of delivering the following power at the various AC input voltages. The AC/DC PSU module shall be able to support short peak power requirements, as stated in the table below.

BBU backup should not be triggered during the peak power loading condition.

The PSU input current waveform shall not be clamped during pulse loading operation.

**Table 5:** Rack power capabilities

| Parameter (min)               | @480V <sub>AC</sub> Vin<br>Option1 | @415V <sub>AC</sub> Vin<br>Option 2 | @400V <sub>AC</sub> Vin<br>Option 3 | Vendors                |
|-------------------------------|------------------------------------|-------------------------------------|-------------------------------------|------------------------|
| Rack average power<br>DC load | 800kW to 1.1MW                     | Up to 1.1MW                         | Up to 718kW                         |                        |
| <2 msec load step             | 175% of TDP                        |                                     |                                     | MSFT<br>Meta           |
| <0.5 msec load step           | 180% of TDP                        |                                     |                                     | MSFT<br>Meta<br>Google |
| <50 msec load step            | 150% of TDP                        |                                     |                                     |                        |
| 1 sec moving average          | 100%                               |                                     |                                     |                        |

### 9.3.3 DC Output – Dynamic loading

The DC output voltage shall not vary by more than specified for undershoot and overshoot with 3ms settling time (with and without capacitive loading on a single unit of 10mF). The measurement shall be from the new steady state value (due to droop) to the voltage spike/dip peaks during a transient load. Voltages are measured at the output of the AC/DC PSU module or/and power shelf rectifier.

BBU backup should not be triggered during the dynamic loading operation. AC input current shall be regulated by PFC.

**Table 6.** Dynamic load requirements.

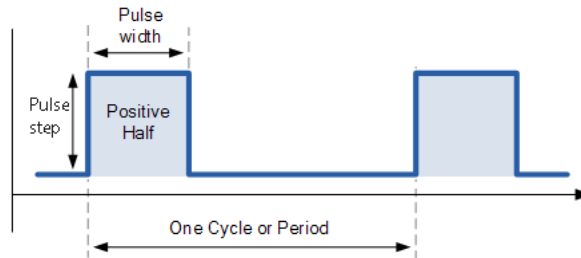
| Step load increase or decrease | Frequency   | Slew rate per PSU | Max undershoot and overshoot |
|--------------------------------|-------------|-------------------|------------------------------|
| 50% step load (10% min. load)  | Up to 10kHz | 1A/us             | ±3%                          |
| 90% step load (10% min. load)  | Up to 10kHz | 1A/us             |                              |
| 140% step load (10% min. load) | Up to 10KHz | 1A/us             |                              |

Dynamic load profiles under test include but not limited to the ones shown in Table 7.

**Table 7.** Example dynamic load profiles.

| No. | Low load | High load | Duty cycle | Frequency                   |
|-----|----------|-----------|------------|-----------------------------|
| 1   | 10%      | 60%       | 50%        | 20Hz, 100Hz,<br>500Hz, 1kHz |
| 2   | 50%      | 100%      | 50%        |                             |
| 3   | 10%      | 100%      | 50%        |                             |
| 4   | 10%      | 150%      | 50%        |                             |
| 5   | 10%      | 60%       | 50%        | 10kHz                       |

|   |     |      |     |        |
|---|-----|------|-----|--------|
| 6 | 50% | 100% | 50% | 10kHz  |
| 7 | 10% | 100% | 50% | 6.5kHz |
| 8 | 10% | 150% | 50% | 4kHz   |



Pulse load profiles under test include but are not limited to the ones shown in Table 8.

**Table 8:** Example pulse load profiles.

| No. | Low load | High load | T_low  | T_high | Average load | Frequency | PSU slew rate |
|-----|----------|-----------|--------|--------|--------------|-----------|---------------|
| 1   | 70%      | 130%      | 0.05ms | 0.05ms | 100%         | 10kHz     | 1A/us         |
| 2   | 20%      | 140%      | 0.1ms  | 0.1ms  | 80%          | 5kHz      | 1A/us         |
| 3   | 40%      | 175%      | 2.5ms  | 2ms    | 100%         | 222Hz     | 1A/us         |
| 4   | 0%       | 80%       | 1ms    | 1ms    | 40%          | 500Hz     | 0.1A/us       |
| 5   | 20%      | 140%      | 50ms   | 50ms   | 80%          | 10Hz      | 1A/us         |
| 6   | 60%      | 140%      | 50ms   | 50ms   | 100%         | 10Hz      | 1A/us         |
| 7   | 30%      | 140%      | 28.6ms | 50ms   | 100%         | 12.7Hz    | 1A/us         |
| 8   | 30%      | 175%      | 0.43ms | 0.4ms  | 100%         | 1.2kHz    | 1A/us         |

The designer shall translate this test performance to system integrator conditions.

#### 9.3.4 End to End Output Power Conversion Efficiency

End to end output power conversion refers to full conversion chain from AC input to DC output, including cooling, distribution, control and energy/backup power losses. Minimum

efficiency values are shown in the table below. AC input is measured at the input to the AC PDU and DC Output is measured at the output of the DC HVDC PDU.

**Table 9:** Total Rack Efficiency

| Parameter  | @480V <sub>AC</sub> Vin | @415V <sub>AC</sub> Vin | @400V <sub>AC</sub> Vin |
|------------|-------------------------|-------------------------|-------------------------|
| @20% load  | >95%                    | > 95%                   | > 95%                   |
| @50% load  | > 98%                   | >97%                    | >97%                    |
| @100% load | >97%                    | >96.5                   | >96.5                   |

## 9.4 Multi Rack Parallel Option

Power rack shall have optional design features that enable paralleling multiple racks to increase power delivery.

Current sharing between racks including droop and active sharing shall be implemented at AC/DC power shelf level or/and rack level.

Synchronization and control features shall be implemented to ensure synchronized power up and power down for the case of multi-rack parallel option.

## 10 Power Rack Building Blocks

### 10.1 AC & DC PDU

The 3 phase AC input and DC  $\pm 400V_{DC}$  HVDC control modules are estimated to take up to 15OU of space and described below.

#### 10.1.1.1 AC PDU Configuration

The AC PDU takes in the AC power from the overhead busway and outputs power to the input of each 3 phase AC/DC PSU rectifier.

The AC PDU shall be designed to:

- Shall accept input connection to quantity up to 12x200A cords
  - Multiple fuses are placed in the PDU
- AC input connection with two-hole compression lugs or connectorized
- Optional AC Panel/Cabinet

The output of the AC PDU 3-phase power is split into smaller rated cables (ex. 200A to 4x50A) cables to be routed to various power rectifier shelves to power the 3-phase PSUs.

#### 10.1.2 $\pm 400\text{VDC}$ Power Distribution

The output of the  $\pm 400\text{V}_{\text{DC}}$  from the power shelves is collected and distributed via a vertical busbar spanning the full height of the rack. The source of the  $\pm 400\text{V}$  is generated by the power shelf or energy storage devices and is transferred via the rack busbar which also is connected to the HVDC Output Protection and Control Modules.

#### 10.1.3 $\pm 400\text{VDC}$ PDU Output Protection and Control Module

The DC PDU Module shall be designed to support up to 400kW output, with built in manageability, output On/Off, protection and telemetry, and no voltage contact on the output cables.

- The HVDC output module shall attach to the  $\pm 400\text{V}$  busbar and contain several output connectors that will transfer the HVDC power to the IT rack via cables
- Multiple output cables per output module, based on two options:
  - Option 1: 50kW per cable, qty as needed to support output power range
  - Option 2: 100kW per cable, qty as needed to support output power range
- The front of the HVDC Output and Control Module shall contain physical on/off switches to turn on/off the corresponding output connector.
- The HVDC Output module shall contain LED indicators, fuses and relays to protect each output connection in case of over current.
- Output voltage disconnect-each output shall disconnect the voltage on the connector before the cable is unplugged from either source or load side and when no output cable is installed.
- Optional active management via microcontroller, for secure access
- Optional module interface with a Rack Management Unit-interface standard defined in chapter 11 (CANBus, Ethernet, etc.)

#### 10.1.4 $\pm 400\text{VDC}$ PE Grounding

There are two options to connect the  $\pm 400\text{Vdc}$  midpoint to PE (Protective Earth) Ground:

- High resistance mid-point grounding PE
- Solid contact mid-point grounding PE

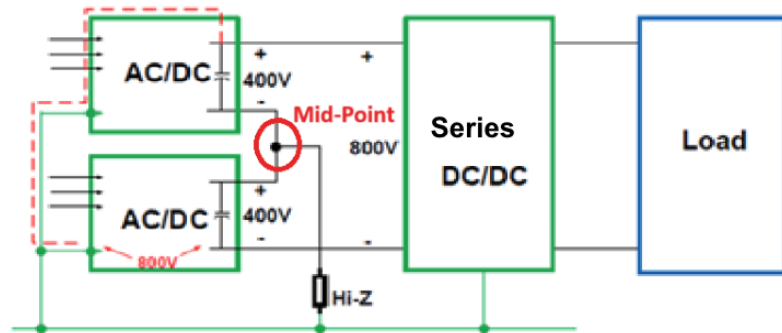
The pros and cons between the two options are addressed in Appendix 1.

Mid-point grounding shall be concealed/covered for no-touch safety

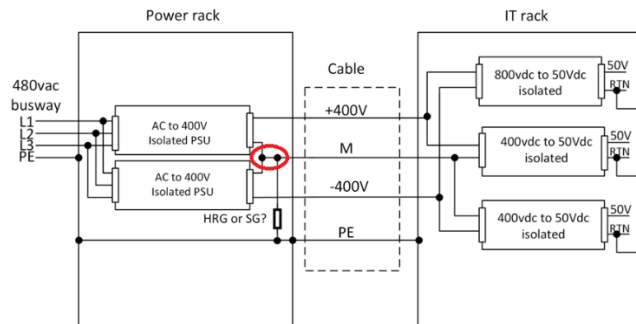
##### 10.1.4.1 High resistance mid-point grounding PE

$\pm 400\text{Vdc}$  routing in the power rack and to IT rack implemented as 3wires+PE as +400Vdc, Midpoint, -400Vdc, and PE.

All outputs are floating voltage design, reference Figure 2 and 3 where mid-point is connected to PE through a High Resistive Midpoint Grounding Scheme. Resistor values is design dependent.



**Figure 2: 2-wire output**



**Figure 3: 3 wire output connection**

#### 10.1.4.1.1 Ground Fault Detector

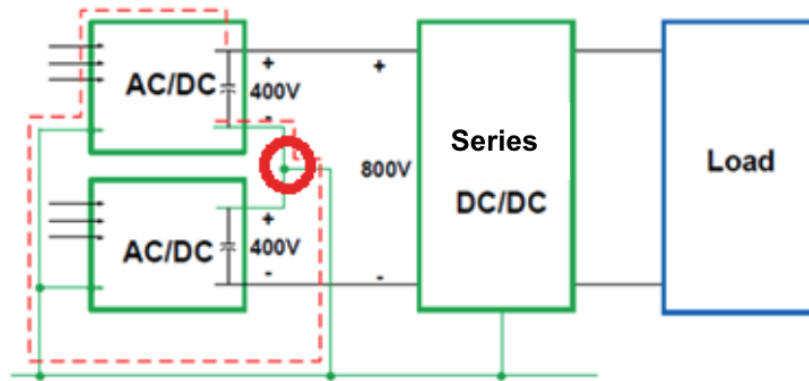
Ground Fault Detector shall be implemented for either solid midpoint grounding or High Resistive Midpoint Grounding implementation.

This building block shall be implemented per each output cable/connection to IT rack. For the case of Midpoint High Resistance Grounding the GFD shall include voltage balance detection circuit.

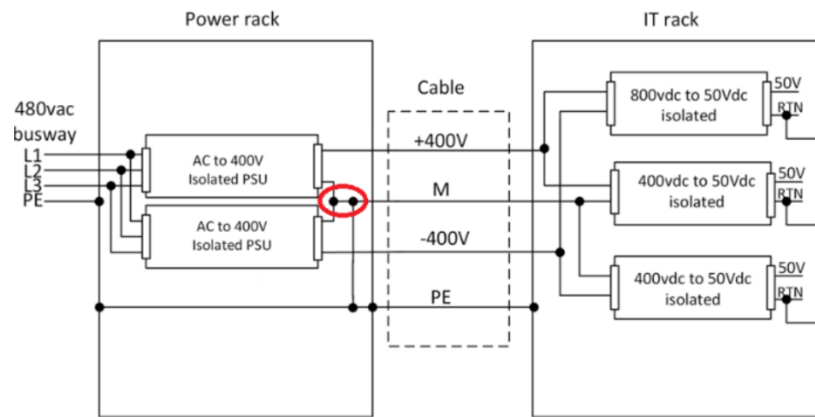
- Optional active management via microcontroller, for secure access
- Optional module interface with a Rack Management Unit-interface standard defined in chapter 11 (CANBus, Ethernet, etc.)

#### 10.1.4.2 Solid contact mid-point grounding PE

Similar 3 wire interface where mid-point is connected as a solid connection, “zero-ohm contact” to PE. All outputs are floating voltage design, reference Figure 4 and 5.



**Figure 4: 2-wire output**



**Figure 5: 3-wire output**

Ground Fault Detector as described in paragraph above is optional in this case.

## 10.2 AC/DC PSU Shelf

Multiple solutions are presented for the AC/DC PSU shelf up to a 20U height, min 21" to 24" wide compatible with rack mechanical specification.

The power shelf can host various numbers of PSUs up to 20U shelf based on form factor and output power requirements.

The power shelf shall be designed as 3-phase AC input and  $\pm 400V_{DC}$  output.

Connectors selected based on PSU requirements.

Design shall have droop and active current sharing between PSUs and Power shelves.

Multiple parallel rack operation defined in section 5.

Design shall support power ranges per electrical spec.

Bus clip connector shall have a thermal sensor attached for thermal shutdown.

- Optional active management via microcontroller, for secure access

- Optional module interface with a Rack Management Unit -interface standard defined in chapter 11 (CANBus, Ethernet, etc.)

### 10.2.1 AC/DC PSU

Currently there are two versions of the AC/DC rectifier modules available in 1OU height form factor. Size information is provided in the table below.

- The quantity of PSUs in a power shelf shall match the number of available output ports on the AC PDU.
- PSU modules are front of rack accessible, hot-swappable, and hot pluggable.
- Complies with electrical requirements in this specification document.
- Galvanic isolated output from AC input and earth/chassis.
- PSU module outputs 400Vdc and is connected at rectifier shelf output as floating output with 3 wires, +400Vdc, -400Vdc and Common/Midpoint/Return
- Design option connected at rectifier shelf output as 2 wires for 800V<sub>DC</sub> . where 800V<sub>DC</sub> and Return shall be safety isolated from PE ground. Downstream converters shall have safety isolated inputs from PE ground.
- Power range per table below

**Table 10:** AC/DC PSU rating and mechanical dimensions

| Parameter                                                    | @480V <sub>AC</sub> Vin                                                                                               | @415V <sub>AC</sub> Vin                                                                                               | @400V <sub>AC</sub> Vin                                                                                               |
|--------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| Rectifier PSU<br>Avg DC load,<br>no<br>redundancy<br>(up to) | <b>Solution 1:</b> 30kW<br><b>Solution 2:</b> 27kW                                                                    | <b>Solution 1:</b> 26kW<br><b>Solution 2:</b> 27kW                                                                    | <b>Solution 1:</b> 25 kW<br><b>Solution 2:</b> 27 kW                                                                  |
| Size XYZ<br>(mm)                                             | <b>Solution 1:</b><br>L810mm*H40mm*W165mm<br>- TBD<br><b>Solution 2:</b><br>L954mm*43mm*W134mm<br>L930mm (w/o handle) | <b>Solution 1:</b><br>L810mm*H40mm*W165mm<br>- TBD<br><b>Solution 2:</b><br>L954mm*43mm*W134mm<br>L930mm (w/o handle) | <b>Solution 1:</b><br>L810mm*H40mm*W165mm<br>- TBD<br><b>Solution 2:</b><br>L954mm*43mm*W134mm<br>L930mm (w/o handle) |

### 10.2.2 AC/DC Power Shelf Control Module

AC/DC power shelf may incorporate a secure control unit, compliant with company specific requirements.

- Optional active management via microcontroller,
- Optional module interface with a Rack Management Unit -interface standard defined in chapter 11 (CANBus, Ethernet, etc.)

Allocated space required per power shelf to fit a control module is listed below.

**Table 11:** Mechanical dimensions for the control module

| Parameter         | X (width) (mm) | Y (height) (mm) | Z (depth) (mm) |
|-------------------|----------------|-----------------|----------------|
| Meta control unit | N/A            | N/A             | N/A            |
| Microsoft RSCM    | 81.2           | 30              | 340.9          |
| Google module     | 42             | 39.7            | 885            |

### 10.3 $\pm 400\text{VDC}$ Distribution Busbar

There shall be a single zone vertical 3 conductor busbar in the rack that distributes +400, -400 and Common. The busbar shall be designed to support up to 1.1MW average load power.

This vertical busbar interconnects with various building blocks such as the AC/DC power shelf, energy storage shelves, HVDC output box, and protection modules.

The busbar is designed such that it shall be concealed/covered for no-touch safety.

The busbar will be air-cooled and shall meet thermal requirements as described in Chapter 8.

### 10.4 $\pm 400\text{VDC}$ to IT rack Distribution Cable

The  $\pm 400\text{V}_{\text{DC}}$  power is brought out of the Diablo 400 rack via multiple HVDC cables.

- Cabled cross links to HVDC/LVDC converter shelf can be a minimum of 63A/50kW up to 125A/100kW each cable.
- $\pm 400\text{Vdc}$  cable connectors on both ends shall have LMFB short pins to prevent arcing
- Each output connector contains at least three power contacts: +400V, Common, -400V; one earth contact and two sense contacts (LMFB).  $\pm 400\text{V}$  and common are all rated from 63A to 125A.
- Connectors shall have a latching mechanism with an integrated physical switch on the output connector, so the cable cannot be disengaged until the switch has turned off the power of the load and the source at the time of disengagement.
- Each output of the  $\pm 400\text{V}$  rail shall be individually current limited according to its maximum current rating.
- Voltage drop budget 0.1% at 5m length for HVDC output cables.
- The respective HVDC cabinet output port shall be de-energized when the HVDC cable is unplugged from either source side or load side. The load side input voltage shall drop below SELV within the safety specified time period.

- HVDC outputs may have the option to be turned off or on through a panel mounted switch located on the HVDC PDU.
- To minimize the number of wires in each HVDC cable set and mitigate concerns about current sharing between cables, a single cable is preferred. However, if more than one cable is used, they should be sized appropriately to accommodate the current sharing tolerance.

## **11 Management, telemetry and communication interface with the sidecar**

### **11.1 Management**

Sidecar/cabinet control module is a dedicated building block that may be used to interface with all Diablo 400 building blocks.

Standard management interface based on Redfish is recommended or may use proprietary interface as desired.

- A. Hyperscaler specific per solution provider
  - May incorporate a secure control unit, compliant with company specific requirements.
    - a. External interface option like Ethernet, etc).
    - b. Internal interface option like CANbus, etc.
- B. Other
  - Use communication ports “as-is”, like building blocks/manufacturer provided interface.

### **11.2 Telemetry**

May use standard interface like Ethernet, I2C, PMBUS, Modbus, CANbus etc.

### **11.3 Communication**

Interfaces may be selected based on the management approach specified above.

- Sidecar to Datacenter network
- Sidecar to sidecar
- Sidecar to IT rack

In-rack communication (building block to sidecar control module) shall be implemented.

## 12 Operating Environmental Conditions

The Diablo 400 power rack and all of its building blocks placed inside the rack is designed to be fully air cooled.

Ambient temperature range: -5°C to 45°C (preferred), -5°C to 35°C (required)

Lower range above, specified as 'required', may be influenced by component temperature ratings and lifetime requirements in sub-systems

Relative humidity: 10% to 90% (non-condensing)

Altitude without de-ratings: up to 3050m (10,000 ft)

### 12.1 Subsystem Thermal Design Requirements

- This section covers sub-systems such as PSU, BBU, CBU, etc.
- Air inlet and outlet temperature difference: Building blocks should be designed to meet a minimum value under a specified load profile/range and environmental conditions which may be specific to each customer.
- Thermal margin: Adequate margins should be reserved to de-rated values on all critical components. In general, a margin of  $\geq 4\%$  of derated temperature rating or  $\geq 3^{\circ}\text{C}$  must be met under rated load and elevated or worst-case environmental conditions. If this requirement cannot be met, failure conditions should be reviewed with the customer. Thermal monitoring: Building blocks shall provide the following parameters via the defined communication protocol. These include (but are not limited to) inlet air temperature, exhaust air temperature, critical component temperatures, fan speed reading(s), fan failure signal(s), etc.

### 12.2 L11/Rack-level Operational Requirements

- Airflow direction: Front-to-back
- Airflow containment: Appropriate containment features such as side panels, etc. should be added to the L11 assembly to ensure all sub-components fit well while mitigating airflow or hot air recirculation from the exhaust side to the cold aisle.
- Optional: Rack (safety) doors: Must be designed or selected to minimize back pressure on sub-systems and this pressure drop must be accounted for as part of the sub-system fan sizing/selection.

## 13 Regulatory and Compliance Requirements

### 13.1 Safety Requirements

Both the fully integrated HVDC power rack system and associated subcomponents/assemblies shall be designed to comply with the safety requirements outlined in the applicable safety standards (e.g. IEC/UL 62368-1, etc.) including all national deviations in addition to the following requirements:

- **SSCR Rating:** AC Mains connected assemblies shall be designed to withstand an available short circuit fault current of 10kA to 40kA, dependent of specific of each application.
  - **Cable assemblies and power cables:** Cables rated above 60Vdc or 42.4Vac shall be UL certified with minimum temperature rating of 90°C and of a type suitable for oil-resistant, damp locations, and hard usage conditions per the NEC (e.g. SJO, SJTOO, SO, SOO, SOOW, STO, W, etc.).
  - **Accessibility:** Components and circuits, including electrical connections intended to be connected/disconnected during either normal or servicing operations while the HVDC power rack system is energized, exceeding PS2 limits shall not be accessible while electrically live or energized. If access to components and circuits exceeding PS2 limits is required for either normal or servicing operations, an interlock, power off switch, or similar mechanism shall be provided for de-energization prior to gaining access.
  - **Creepage and Clearance:** Design of Sidecar modules and connectors must comply with applicable safety standards for spacing requirements, e.g., IEC 62368. The designer/implementer may use discretion to exceed the standard requirements when appropriate for their use case. Use cases shall include AC input and DC output parameters, across thermal and environmental conditions, including peak potential, transient or test conditions.
- 
- **Overcurrent Protection:** HVDC outputs shall be provided with overcurrent protection.
  - **Protective earthing and bonding:** HVDC cables shall be provided with dedicated protective grounding for connections between both the Power and IT Racks to ensure a safe and reliable ground for fault currents. All protective grounding terminals shall be securely bonded and marked in compliance with IEC 60417 (ref. no. 5019).
  - **Leakage current:** The HVDC power rack system shall not have a leakage current exceeding 3.5mA (AC) and 10mA (DC).
  - **Fault Detection:** Ground Fault Detection features shall be provided for effective protection against fault conditions.
  - **Stability:** The HVDC power rack system shall remain stable and not tip over when tilted up to 10° in any direction.
  - **Energy Storage Systems (ESS):** Energy Storage Systems (ESS), including battery modules and cells, shall be designed to comply with the safety requirements outlined in all the applicable safety standards (e.g. UL 9540, UL 9540A, UL 1973, IEC 62133, etc.), NFPA 855, and transportation requirements (e.g. UN 38.3, IATA DGR, IMO International Maritime Dangerous Goods Code, ect.).

Fire propagation tests per UL 9540A shall be performed at the cell, module, and rack levels. No external flames are permitted during rack-level testing and shall comply with the criteria not to trigger the installation base fire tests.

## 13.2 Electromagnetic Compatibility (EMC) Requirements

The fully integrated HVDC power rack system and associated subcomponents/assemblies shall be designed when integrated together to comply with requirements for Class A devices when tested according to CISPR 32 and CISPR 35 standards, and applicable national regulations (i.e. FCC CFR 47, part 15 in the USA, ICES-003 in Canada, EN 55032 and EN 55035 in Europe, etc.) across the full range of supported operations and configurations (i.e. idle, normal, minimum/maximum workloads, etc.) in addition to the following requirements:

| Test                                              | Test Method         | Requirement                                                                                                                                                  |
|---------------------------------------------------|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Radiated and Conducted Emissions                  | ANSI C63.4/CISPR 32 | Min -6 dB margin                                                                                                                                             |
| Electrostatic Discharge (ESD) - Contact Discharge | EN 61000-4-2        | Preferred $\pm 5.6$ kV, Criteria A, /min $\pm 4$ kV, Criteria A,                                                                                             |
| Electrostatic Discharge (ESD) - Air Discharge     | EN 61000-4-2        | Preferred $\pm 11.2$ kV, Criteria A /min $\pm 8$ kV, Criteria A                                                                                              |
| Surge                                             | EN 61000-4-5        | Differential Mode: Preferred $\pm 2$ kV, Criteria A /min $\pm 1$ kV, Criteria A<br>Common Mode: Preferred $\pm 4$ kV, Criteria A /min $\pm 2$ kV, Criteria A |
| Radiated Immunity                                 | EN 61000-4-3        | Min 10V/m Criteria A                                                                                                                                         |

## 13.3 Environmental, Eco-Design, and Sustainability Requirements

The fully integrated HVDC power rack system, associated subcomponents/assemblies, and any packaging shall be designed to comply with worldwide environmental, eco-design, and energy efficiency laws and regulations (e.g. RoHS Directive 2015/863/EU, REACH Regulation (EC) No 1907/2006, WEEE Directive 2012/19/EU, SCIP and EPA 40 CFR751, Commission Regulation (EU) 2021/341, etc.).

## 14 Appendix A

### 14.1 High Resistance Grounding (HRG)

- No arcing or fire risk when +/-400V short to PE
- No serious injury to human body if touching +/-400V due to limited leakage current
- ⊗ Increased spacing requirements of downstream DCDC -to be confirmed by DCDC Designer/safety.
- ⊗ Complex design and maintenance

Pro1: No electric shock risk if touching +400 V or - 400 V with reference to ground.

Electric shock risk still exists between +400 to -400.

Pro2: Mitigates arcs due to a line to ground fault. Note: does not protect against line-to-line faults

Con1: Electric shock risk if touching +400 V or - 400 V under single fault condition where the other leg faults to ground

Con2: Fault detection is required to effectively resolve a fault to ground before a second fault occurs.

Con3: Putting fault detection at payload rack tray level may be expensive. Having fault detection on per rack level (or even Sidecar level) will be difficult to troubleshoot.

Con4: First fault to ground will not disrupt operation and therefore is at risk of being a lower priority to resolve. Second fault to ground can create a fire/electric shock risk.

Con5: Sensors must be calibrated and tested on a recurring basis. A defective sensor must be replaced immediately. If the sensors are not operating properly this could give you a false sense of security.

Con6: Multiple sources in parallel would reduce the impedance to ground and increase the touch current

Con7: A component in a payload rack that was designed for high impedance grounding scheme may not be suitable for use in a low impedance grounding scheme. If the component is deployed in a Data Center with HVDC bus but it does not employ a high impedance grounding scheme, careful review of the component must be done to ensure compatibility. For example, a component designed for use in a high impedance grounding system may not have input fuses or functional insulation may have been used between the 400V to chassis.

### 14.2 Solid Grounding (SG)

- Simple design and DC maintenance
- Safety distance based on 400Vdc.
- ⊗ Blown fuse or burnt component when +/-400V short to PE

⊗ In certain configurations, grounding current may need to be mitigated to prevent potential issues.

Pro1: Widely used solution for AC power distribution.

Pro2: Fuse instead of sensors-based circuit protection to cut power on a fault condition

Pro3: Easier to identify the problem circuit

Pro4: Safety standards have clear requirements for products used on a system with low impedance grounding

Con1: HV parts shall not be accessible, as high body current discharge possibility exists

Con2: Arcing may occur until fuse operates

Con3: The bond between the accessible conductive part and ground must be able to carry fault current until overcurrent protection opens.

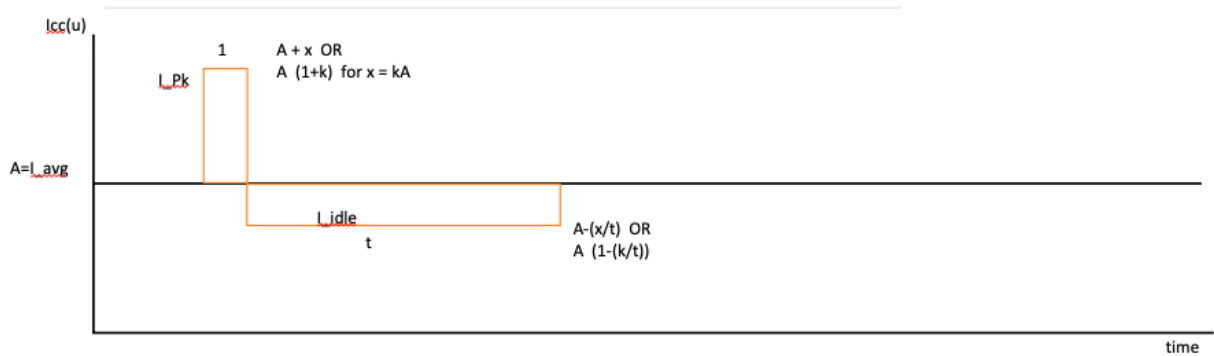
Con4: multiple power racks with solid grounding generate multiple ground loops

## 15 Appendix B

### 15.1 Average current vs RMS current thermal impact

Let  $I_{cc}(u)$  and  $V_{cc}(u)$  be Random Variables, able to take any value from a universe of values ( $u$ ).

Let function  $E\{ \}$  be Expected Value, i.e., average, of a given Random Variable being operated on.



$$A = E\{I_{cc}(u) \text{ at GF}\} = I_{avg}$$

$$A+x = I_{Pk} = E\{I_{cc}(u) \text{ Worst Case Max}\}$$

$$I_{rms}^2 = \frac{1(A+x)^2 + t\left(A - \frac{x}{t}\right)^2}{1+t}$$

$$\frac{I_{rms}}{I_{avg}} = SQRT\left(1 + \frac{k^2}{t}\right) \text{ for } x = kA$$

Implementations of GPUs CPUs Average (Thermal) Power algorithms could measure and average the processor power. Within the processor, the dominant power is the Dynamic Power which is linear with Current, i.e.,  $P_{dyn} = I_{dyn} * V_{cc}$ , where  $I_{dyn} = C * V_{cc} * \text{Freq.}$ , the controlled and measured Processor Average Power will render a resultant Average Current proportional to Average Power.

On the other hand, power losses in hardware circuitry (such as VRs) are result of current over Real Resistance. This VR average power calculation requires use of  $I_{rms}$  (see below).

$I_{RMS}$  to  $I_{AVG}$  ratio is the VR steady state thermal "gain" or how much thermally bigger the VR needs to be when compared to one just designed to  $I_{AVG}$  rendered by processor at TDP.

VR thermal gain is dependent on the magnitude of  $I_{Pk}$ , **i.e., k!!**... the higher the  $I_{Pk}$ , the higher the impact on the VR.

You need Average Power to determine thermal power (you get joules out of  $P_{avg}$ )

You need RMS current to determine Average Power at the VR ckt.

$$P_{inst} = I_{cc}(u)^2 * R$$

$$P_{avg} = E\{P_{inst}\} = E\{I_{cc}(u)^2 R\}$$

$$P_{avg} = E\{I_{cc}(u)^2\} R$$

$$P_{avg} = (I_{RMS})^2 R = \frac{(V_{RMS})^2}{R} = I_{RMS} V_{RMS}$$